Orchestrating Microbiome Analysis with Bioconductor

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Abstract

The expansion of microbiome research has led to the accumulation of interlinked datasets encompassing versatile taxonomic and functional assays. The analysis of increasingly large and heterogeneous multi-modal microbiome data would benefit from unified approaches supporting the design of modular data science workflows through interoperable methods. Building upon the optimized statistical programming framework for multi-assay data integration recently developed by the Bioconductor project, we introduce a community-developed open source framework for microbiome data science. In contrast to the previous alternatives, the methodology is specifically designed to support joint analysis of hierarchical, interlinked, and heterogeneous multi-table datasets that are increasingly common in modern microbiome research. This framework encompasses open data, methods, tutorials, and an active online community, thereby supporting standardized and reproducible data wrangling, joint analysis, and reporting. We have detailed the functionality and usage in the OMA book (https://bioconductor.org/books/release/OMA), which offers guidance for prospective users and contributors.

Keywords: microbiome, Bioconductor, data science, data integration

Introduction

Modern data science applications critically rely on open-source methods created by the research community[1;2]. Open software ecosystems have become central innovation hubs, driving technological and cultural transformation[3]. The Bioconductor project has become a significant distribution channel for open data science methods in the life sciences. Its vast developer community maintains over 2,300 quality-controlled R packages for various fields of life science informatics[4]. Additionally, Bioconductor has evolved into a global community that supports data science collaboration and training[5], playing an increasingly important role in advancing computational skills that are essential for modern microbiology education[6].

Microbiome research focuses on how microbial communities vary, function and interact with their environment and a potential host[7]. The technologies used in this field generate vast amounts of high-dimensional and hierarchically structured data, commonly in the form of microbial DNA sequences detected in a set of samples. Following the rapid expansion of microbiome research and technological advances, metagenomic sequencing is increasingly complemented by other types of high-throughput measurements. This has led to the accumulation of multi-modal datasets—collections that combine different types of molecular data. These interlinked datasets encompass diverse taxonomic and functional assays, ranging from microbial abundances and phylogenetic relationships to functional profiles.

Microbiome data scientists are thus increasingly facing the practical challenges of integrating heterogeneous data across multiple data modes into reproducible workflows. The unique statistical properties and complexity of microbiome data necessitate specialized approaches in their analysis[7–9], yet the diversity of proposed solutions and inconsistent outcomes[10–12] can make it difficult to identify the optimal methods and build interoperable workflows[13;14]. Accordingly, a need for standardization has been highlighted[7]. Open microbiome data science *frameworks* have emerged to provide sets of interoperable methodologies in specific computational environments[15–18], and a broader ecosystem of digital resources, engaged user communities and collaboration culture that extend the capacities far beyond the underlying technical framework[19]. However, frameworks focused solely on microbiome data science fall short on addressing the need to integrate data across multiple modalities, such as metagenomics, metabolomics, and host genomics. The Bioconductor ecosytem integrates the methods of microbiome data science into a wider statistical programming framework, where interoperable methods and workflows for single-cell omics[20], spatial transcriptomics[21], metabolomics[22], proteomics[22;23], and other fields are concurrently developed.

Here, we introduce a stable and optimized data science ecosystem supporting the integration and analysis of multi-modal datasets in microbiome research. It provides harmonized data import and representation, and fast, robust methods for transforming abundance tables into biological insights. These methods have been extensively tested and distributed through R/Bioconductor packages (https://bioconductor.org/packages), supporting the community-driven development of the framework and integration within the broader Bioconductor ecosystem. To promote evidence-based best practices, we introduce the online book *Orchestrating Microbiome Analysis with Bioconductor* (OMA), which represents a comprehensive compilation of established approaches and emerging trends in microbiome data science using the R/Bioconductor ecosystem.

Data infrastructure

Data scientists routinely deal with interlinked assays, hierarchical data (*e.g.*, ontologies, trees), relational data (*e.g.*, networks), and supporting side information. This side information can include sample attributes such as treatment group, collection site, or time point. The ability to integrate heterogeneous data elements is essential in biological research, where the individual components of a system can seldom be studied in isolation. However, this growing complexity introduces additional overhead, as researchers must track samples and features across multiple tables and manage an increasing number of data elements and non-trivial links between them. Moreover, input data formats vary across different methods, leading to technical complications and time lost on data wrangling. Bioconductor is addressing these challenges through standardized data structures that are supported by an ecosystem of quality-controlled, interoperable methods. This allows the analyst to invest more time in core analytical tasks.

Bioconductor data structures support statistical analysis of complex data combinations[24] and are widely adopted by various research communities. They have been implemented through specialized object-oriented classes that integrate multiple data elements into a single structure known as a *data container*[25]. The SummarizedExperiment (SE) class[24;26] is the primary data container underlying the Bioconductor ecosystem (Figure 2 a and Table 1). Its position among the top 1% most-downloaded packages in Bioconductor reflects its widespread recognition and increasing adoption among developers (Figure 1). SE provides a standardized solution to store tabular data by linking numeric abundance matrices with side information on features or rows (*e.g.*, taxonomy), and samples or columns (*e.g.*, origin, collection time, host health).

The interoperability of SE with other Bioconductor data structures enables the transfer of methods across different fields of omics research, thereby promoting the development of modular and scalable data science methods and ultimately yielding improved overall quality control and user support[5]. In particular, the SE framework has been extended to similar data integration tasks, notably in single-cell and spatial omics through the SingleCellExperiment (SCE)[20] and SpatialExperiment[21] classes.

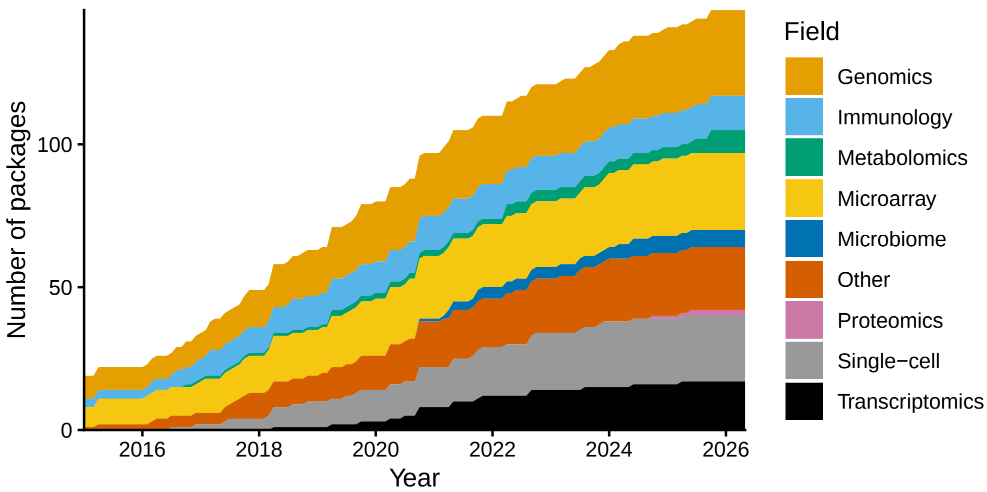

Figure 1: **Adoption of the SummarizedExperiment (SE) data container among Bioconductor developers.** The growth in the number of Bioconductor packages supporting SE over time is shown, categorized by their respective fields. Packages are counted by the date they were added to Bioconductor. Some packages added the SE support after their original release.

Hierarchical data structures. Microbiome data scientists frequently need to integrate information on feature phylogenies and sample hierarchies to accurately reflect fine-scale variability in microbiome data. While maintaining interoperability with the broader Bioconductor ecosystem, the TreeSummarizedExperiment (TreeSE) class[27] extends the SE and SCE data structures to hierarchical datasets by incorporating both feature and sample organizations as tree structures (Figure 2 b and Table 1). Moreover, TreeSE supports the storage of other common elements of microbiome data, for example, reference sequences, and incorporates results from common analytical procedures such as dimensionality reduction, statistical transformations, and agglomeration by taxonomic, functional or other information. The interoperability with other Bioconductor classes facilitates scalable analysis and the design of modular yet flexible workflows, as TreeSE inherits full compatibility with methods designed for SE and SCE. In particular, the direct interoperability with the widely adopted SE data science ecosystem distinguishes the (Tree)SE framework from phyloseq, which is another popular Bioconductor data container initially developed with a focus on 16S amplicon data analysis[18]. In contrast, TreeSE provides a more general framework, as it not only accommodates the typical components of a phyloseq object (taxa abundances, sample metadata, phylogenetic tree, and reference sequences), but also extends to multi-modal datasets encompassing parallel taxonomic levels, functional predictions, and resistome, metabolome or other profiles and alternative feature representations.

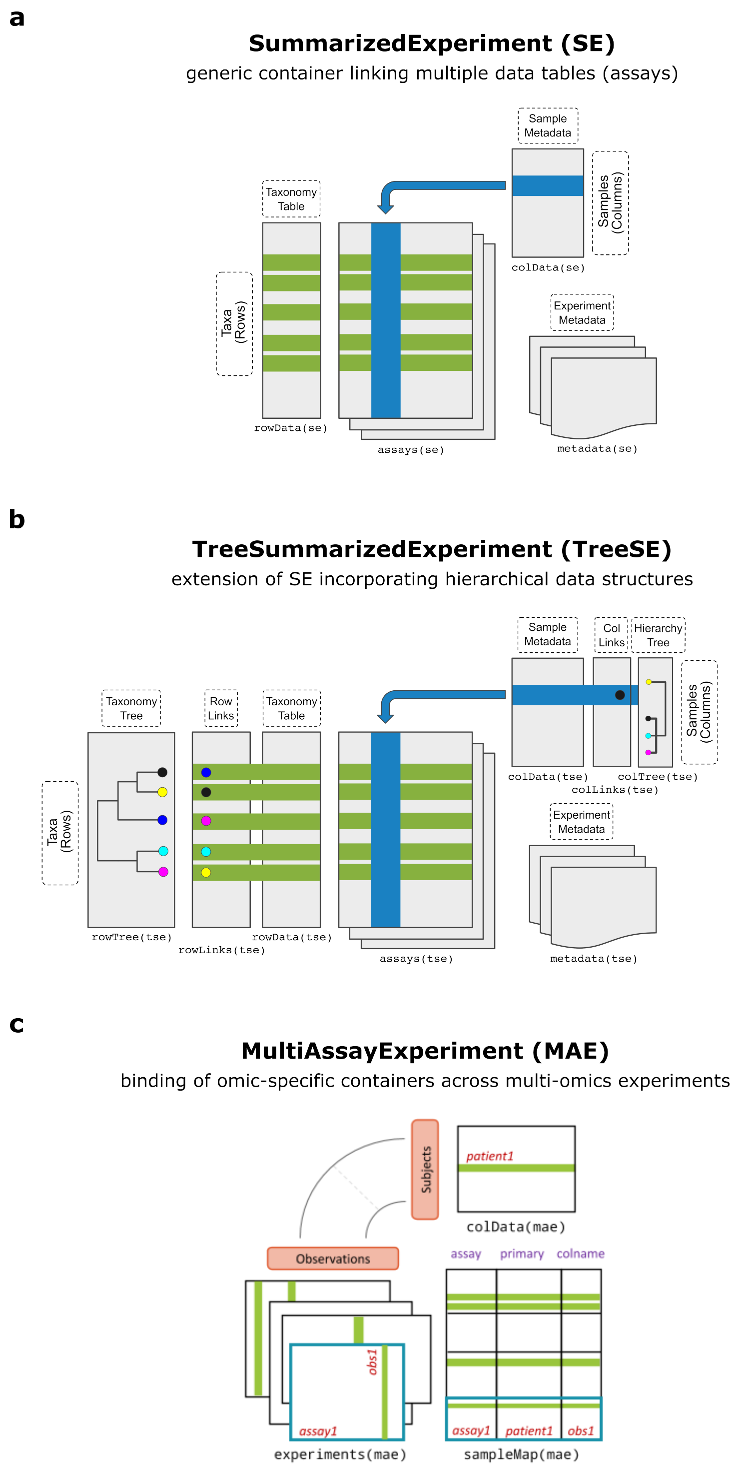
a
SummarizedExperiment (SE)
generic container linking multiple data tables (assays)
Sample Metadata
Samples (Columns)
Taxonomy Table
Taxa (Rows)
colData(se)
rowData(se)
assays(se)
Experiment Metadata
metadata(se)
b
TreeSummarizedExperiment (TreeSE)
extension of SE incorporating hierarchical data structures
Sample Metadata
Col Links
Hierarchy Tree
Samples (Columns)
Taxonomy Tree
Row Links
Taxonomy Table
Taxa (Rows)
colData(tse)
colTree(tse)
colLinks(tse)
Experiment Metadata
rowTree(tse)
rowLinks(tse)
rowData(tse)
assays(tse)
metadata(tse)
c
MultiAssayExperiment (MAE)
binding of omic-specific containers across multi-omics experiments
Subjects
patient1
colData(mae)
Observations
assay
primary
colname
obs1
assay1
assay1
patient1
obs1
experiments(mae)
sampleMap(mae)

Figure 2: **Integrative Bioconductor containers for microbiome data. (a) SummarizedExperiment (SE)**[24] is a generic container for single-omics experiments. It accommodates *assays* that describe the feature abundances in each sample (*e.g.*, taxonomic units, resistance genes, or predicted functions), and tabular side information on each sample (*colData*) and feature (*rowData*), *e.g.*, the taxonomic classification. **(b) TreeSummarizedExperiment (TreeSE)**[27] supports hierarchical, multi-table microbiome analysis. In addition to the SE elements, TreeSE incorporates tree information on feature hierarchies, such as phylogenetic relations between the taxonomic units (*rowTree*) or host species (*colTree*), and results of dimensionality reduction (*reducedDims*). TreeSE can thus link multiple abundance tables and their transformed versions (*e.g.*, statistical transformation, taxonomic agglomeration, or dimensionality reduction) with tabular and hierarchical side information on the features (rows) and samples (columns). Additional slots for reference sequences and experiment metadata are available. **(c) MultiAssayExperiment (MAE)**[28] facilitates the integration of interlinked datasets representing different omics modalities linked by potentially non-trivial sample mappings. Reproduced from[28].

Multi-assay data structures. Contemporary microbiome research frequently involves data across multiple measurement modalities for enhanced functional and mechanistic insights. This brings the need to find an appropriate representation for each data type and map the links between them. Whereas the TreeSE data container is broadly applicable to different modalities such as taxonomic, resistome, transcriptome, proteome, and metabolome profiling data, dedicated data containers have been developed for certain modalities by the Bioconductor community to address their specific characteristics; examples include single cell sequencing[20] and host genomics[29]. Linking data across modalities presents an additional technical challenge, as multi-omics datasets are often sparse and only a subset of the samples may be shared across multiple modalities[30]. In other cases, a single sample in one modality may link to several samples in another modality[31] (*e.g.*, due to technical or spatial replication). These scenarios require the ability to integrate multiple data modalities into a single object and create flexible sample mappings across them. The MultiAssayExperiment (MAE) data container[28] addresses these challenges by introducing a formal mapping layer (sampleMap) that links biological samples to their corresponding measurements in each modality (Figure 2 c and Table 1). This design supports one-to-one, one-to-many, and partially overlapping sample relationships across experiments. Importantly, this mapping is also accessible for downstream operations. For instance, when subsetting a MAE object (*e.g.*, selecting samples with both microbiome and metabolome data), the framework automatically resolves the intersection of available samples and ensures consistent alignment across all modalities. Such structured handling of sample relationships is particularly important for integrative methods, as it reduces the risk of sample mismatches, a common source of irreproducibility in ad hoc multi-table analyses. For instance, diagonal integration methods[32;33]—where shared features across data types may not be explicitly defined—can benefit from such a systematic framework that enforces anchor alignment. This reflects lessons learned from single-cell data integration, where shared anchors and hierarchical mapping have proven essential for interpretability. To summarize, flexible mapping of samples across several data objects is key for efficient data integration, method sharing, and overall interoperability.

Table 1: Key elements of the Bioconductor data containers for microbiome analysis.

Slot	Content	SE	TreeSE	MAE[a]
Tables				
assays	tables of features (rows) and samples (cols)	X	X	X
altExps[b]	experiments with variable number of features		X	X
reducedDims	results of dimensionality reduction		X	X
Rows				
rowData	side information on features	X	X	X
rowPairs	linkage between pairs of associated features		X	X
rowTrees	hierarchical organization of features		X	X
referenceSeq	reference sequences of features		X	X
Columns				
colData	side information on samples	X	X	X
colTrees	hierarchical organization of samples		X	X
sampleMap[c]	flexible sample mapping across experiments			X
Other				
metadata	additional information on the experiment	X	X	X

[a]While MAE does not contain conventional slots (with the exception of colData), its experiments can include any number of data containers that provide those slots. [b]altExp samples must match assay samples in a 1-to-1 mapping. [c]The sampleMap supports diverse mappings between samples from different experiments (1-to-1, 1-to-many, many-to-1 and many-to-many). SE, SummarizedExperiment; TreeSE, TreeSummarizedExperiment; MAE, MultiAssayExperiment.

Data resources. Several open microbiome data resources are supported, enabling direct data import into the aforementioned data containers for further downstream analysis within the Bioconductor methods ecosystem (Table 2). Examples of the available data resources include curatedMetagenomicData[34], HoloFood[35], microbiomeDataSets[36], and the EBI/MGnify database[37]. In addition, Bioconductor packages often include built-in demonstration datasets. Collectively, these resources encompass taxonomic and functional profiles from thousands of published microbiome studies and tens of thousands samples that can be accessed with the available tools for research and educational purposes.

Table 2: Open microbiome data resources (last accessed on 05/07/2026).

Data resource	References	# Studies	# Samples
MGnifyR	[37;38]	5,616	429,108
Metalog	[39]	1,074	157,483
curatedMetagenomicData	[34]	93	22,588
miaData	[40]	9	20,404

Data resource	References	# Studies	# Samples
microbiomeDataSets	[36]	6	19,100
HMP2data	[41]	3	11,493
HoloFoodR[a]	[35;42]	9	9,990
MicrobiomeBenchmarkData	[12]	6	1,125

[a]HoloFood microbiome datasets are deposited in the MGnify database. HMP, Human Microbiome Project.

Alternative frameworks

The (Tree)SE framework is one of several approaches to microbiome analysis. Common alternatives include phyloseq[18]—with its extension speedyseq—, which is also part of Bioconductor, and the command-line toolkits Mothur[16], QIIME 2[17], and Anvi'o[15]. Each of these frameworks was designed to tackle recurring challenges in microbiome data science—a field where no single solution yet provides the most complete, coherent, and efficient approach.

Compared to its alternatives, the (Tree)SE framework relies on a more compact, efficient data structure that is fully interoperable with the larger Bioconductor ecosystem. This way, it provides an evidence-based, accessible solution for developing end-to-end microbiome analysis workflows that can integrate concurrent Bioconductor development efforts, without being limited exclusively to methods for microbiome data analysis.

This integrative approach to bioinformatics is also found in the Python library scikit-bio, which provides general methods for various omics modalities[43]. While relatively efficient, its infrastructure for microbiome analysis is not yet as extensive and does not leverage integrative data containers such as TreeSE. Nonetheless, certain scikit-bio modules for community diversity estimation are available through the larger QIIME 2 framework for microbiome analysis.

Based on an extensive benchmark of routine operations in microbiome analysis applied to subsets of the Metalog database[39], the (Tree)SE framework shows substantial performance gains over alternatives (Figure 3 and Supplementary Figure 1). While speedyseq performs faster in certain operations, TreeSE tends to scale more efficiently in terms of execution time and allocated memory as feature and sample sizes increase. As most operations exceeded resource limits for very large sample subsets, future development will need to focus on further optimizing these routine operations in the (Tree)SE framework, *e.g.*, by executing costly computations in C++ and by leveraging the built-in support for sparse and delayed matrices in SE-based data containers, which can significantly improve method efficiency[24]. To facilitate migration from alternative frameworks, the OMA book provides conversion tables and utility functions to convert between TreeSE and external formats.

Data preparation

While several tools for microbiome data analysis are available[8;9;45], their implementations typically differ in terms of expected input and output formats. Common data

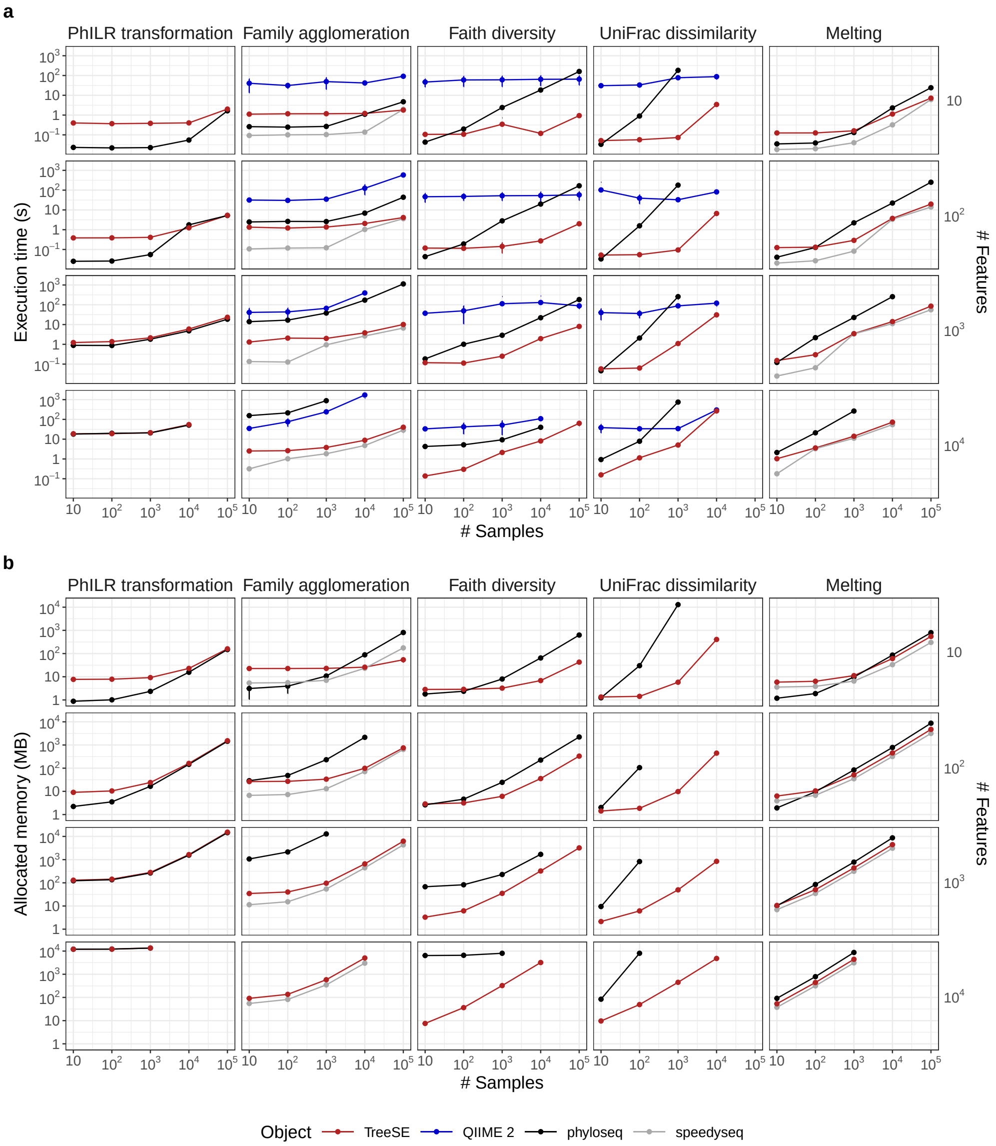

Figure 3: **Execution time and allocated memory for routine microbiome data operations.** TreeSummarizedExperiment, QIIME 2 and phyloseq—with its extension speedyseq—are three popular frameworks for microbiome analysis. We benchmarked them for five routine operations applied to random subsets of features (rows) and samples (columns) from the Metalog database[39]. Both execution time (s) and allocated memory (MB) were measured as the mean value over ten replicates with varying subsets of the same dimension. Errorbars show the 95% confidence interval of the performance estimate. The benchmarking was conducted in R using the bench library with 4 CPU cores and 16 GB RAM. QIIME 2 lacks allocated memory estimates because memory cannot be monitored for external processes. The source code for the benchmarking analysis is openly available[44].

standards promote modular data science workflows, speeding up methods development, benchmarking, and replication. Building upon these principles, we describe a set of interoperable software libraries that support microbiome data integration and analysis based on the two integrative data containers, TreeSE and MAE (Figure 4).

This framework specifically supports the downstream analysis of high-throughput sequencing data from microbiome experiments after summarizing the original measurements into abundance tables (*e.g.*, taxonomic or functional profiles). Taxonomic abundance tables are one of the most commonly encountered data types. Common measurement platforms include amplicon-based marker gene studies (*e.g.*, 16S rRNA) and metagenomic sequencing, but alternative profiling techniques are available including phylogenetic microarrays[46] and high-density optical mapping[47]. The interoperability of the data containers support modular workflow design for integrating different omics data types into the broader Bioconductor ecosystem. This mitigates reproducibility challenges posed by rapidly evolving reference databases and profiling tools (*e.g.*, MetaPhlAn) by enabling structured coexistence and direct comparison within a unified, coherent framework[48;49].

Data import. While the TreeSE data container can be populated with microbiome data from common tabular data types, certain non-standard file formats may require additional data wrangling. To streamline data import, we provide importers for a variety of standard microbiome file formats including BIOM[50], MetaPhlAn/HUMAnN (MetaPhlAn v2–4, HUMAnN v3)[51], Mothur[16], QIIME 2[17] and the TAXonomic Profile Aggregation and standardization format (taxpasta)[52]. These importers bridge the gap between the raw sequence processing and downstream statistical analysis in R, allowing researchers to focus on interpreting biological patterns rather than wrangling file formats. Moreover, converters between alternative data structures in R are available supporting seamless conversions between TreeSE and BIOM, DADA2[53] and phyloseq (the mia::convert* functions). Similarly, microbiome data stored in TreeSE objects can also be exported to external services such as MicrobiomeDB[54] and other languages (*e.g.*, Julia). These converters facilitate widespread access to microbiome analysis methods. Imported datasets from different omics modalities can then be interlinked within the MAE data container with its flexible sample mappings as described above. This integrated data object can then be analyzed with dedicated downstream methods.

Quality control and exploration. After importing the data into R, the recommended first step involves basic data exploration and quality control[55]. In addition to methods for identifying contaminant sequences and calculating summary statistics, the framework includes a versatile set of custom visualization methods to assess data variability, outliers, homogeneity of variance and other aspects that may need to be considered in downstream data analysis. The mia and miaViz packages provide convenient access to many operations for the integrative data containers regarding common tasks in microbiome data exploration and visualization (Figure 5).

Filtering and agglomeration. Microbiome datasets can contain tens of thousands of unique features depending on the sequencing resolution, used reference databases, and the scope of analysis. Moreover, the integration of multiple omics modalities adds another level of challenges. To summarize data into biologically meaningful groups or to reduce multiple comparisons the data is often subsetted, filtered or agglomerated to higher-level taxonomic or functional categories or stratified by sample groups. The package ecosystem, and the mia package in particular, feature various tools for such tasks. For instance, the prevalence-based methods can be helpful to remove rare features that are likely sequencing artifacts or present otherwise limited value for the analysis. The mechanism of alternative experiments (altExp) allows agglomeration by taxonomic levels or other features such as pathogenicity or functional potential (*e.g.*, butyrate production[56], gut-brain modules[57], and other host-associated signatures[58]). The agglomerated data can be incorporated as a new alternative experiment within the original data object. The automated tracking of feature and sample links ensures that changes are propagated across all relevant elements of the data object, such as transformed abundance tables, phylogenetic trees, or interlinked omics modalities. This helps to avoid redundant processing and enables efficient management of interlinked multi-modal datasets.

Transformation. Statistical transformations commonly used in microbiome research include relative abundance[59], tree-aware phylogenetic ILR (phILR)[60], centered log-ratio (CLR), and robust CLR (rCLR). The latter three aim to mitigate compositionality bias[61]. A related concept that controls for unequal sampling depth is data rarefaction. Rarefaction is the process of repeatedly subsampling data to equal read depth and averaging computed metrics[62]. Subsampling data to equal read depth (*rarefying*) without iteration is not recommended[63–65]. Our framework supports these and a variety of other transformations commonly applied to microbiome data.

Data enrichment. Incorporating additional data, such as transformed feature abundance tables, is a frequent operation in applied data analysis, and it often takes place iteratively. The integrative data containers support gradual and organized accumulation of interlinked data elements including transformations, aggregated data, reduced dimensions, and derived data on samples and features. We generally recommend initializing the data objects at the beginning of the workflow, where they can then be gradually enriched as the project advances. Data can be enriched by importing new external information as well as by deriving new information from the data itself into the same data container as described in the next section. By systematically extending the upstream data preparation step—instead of ad-hoc calculations scattered amidst the workflow—one can achieve a well-organized and transparent provenance of the integrated data object.

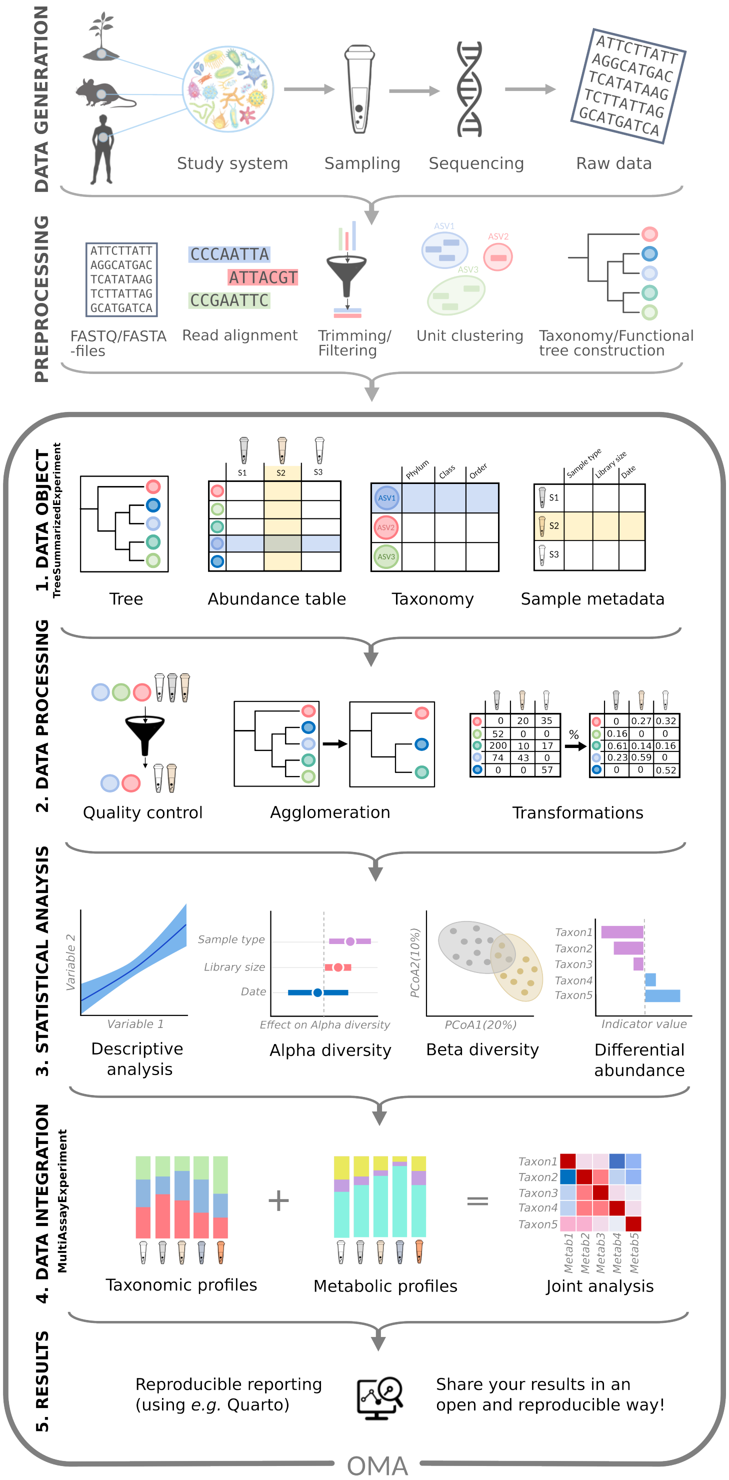
DATA GENERATION
Study system
Sampling
Sequencing
Raw data
PREPROCESSING
FASTQ/FASTA -files
Read alignment
Trimming/ Filtering
Unit clustering
Taxonomy/Functional tree construction
1. DATA OBJECT
TreeSummarizedExperiment
Tree
Abundance table
Taxonomy
Sample metadata
2. DATA PROCESSING
Quality control
Agglomeration
Transformations
3. STATISTICAL ANALYSIS
Variable 2
Variable 1
Descriptive analysis
Sample type
Library size
Date
Effect on Alpha diversity
Alpha diversity
PCoA2(10%)
PCoA1(20%)
Beta diversity
Taxon1
Taxon2
Taxon3
Taxon4
Taxon5
Indicator value
Differential abundance
4. DATA INTEGRATION
MultiAssayExperiment
Taxonomic profiles
Metabolic profiles
Joint analysis
5. RESULTS
Reproducible reporting (using e.g. Quarto)
Share your results in an open and reproducible way!
OMA

Figure 4: **Multi-modal data integration workflow.** Following data generation and preprocessing (top two panels), the integrative framework covers five steps of microbiome data integration and analysis: 1) Data is wrapped into a TreeSE data object (Figure **??**), containing information on the samples (tube icons) and their features (*e.g.*, taxonomic, functional, or resistome profiles; colored circles). TreeSE objects can accommodate multiple assay tables as well as additional hierarchical structures. 2) The TreeSE data object can then be processed to get cleaner data. This can include filtering out low-quality samples and features, transforming assay data for example by normalizing it to enhance the comparability of sample-specific abundances, or agglomerating the feature data to specific taxonomic levels or functional categories. 3) The cleaned data can then be analyzed using statistical models to uncover relations among samples (*e.g.*, with ordination methods), analyze drivers of variation in community diversity and composition or test for differential abundance (*e.g.* between sampling time or location). 4) The framework supports the integration of various types of omics modalities through MAE and omic-specific data containers provided by Bioconductor. Here, for example, the taxonomic and metabolomic profiles of samples can be summarized and compared with joint analysis among features. 5) Finally, Quarto notebooks can aid reproducible analysis and reporting.

Downstream data analysis

This framework is designed to improve the management of integrative analysis workflows. It facilitates joint operations on interlinked data elements, helping to shift the focus towards multi-modal analyses and methods development while also supporting standard data analyses in microbiome research.

Microbiome data analysis often relies on specialized statistical approaches, since data sparsity, zero-inflation, compositionality, and heteroscedasticity pose challenges for standard methods[8;59]. Furthermore, microbiome data is hierarchically structured across taxonomic, ecological, temporal, and spatial levels[7;66]. Alternative processing pipelines introduce different biases and generate varying data formats; for example, the same sample can yield differing taxonomic profiles depending on the computational method used[67]. These and other unique characteristics have wide-reaching implications for microbiome analysis, necessitating dedicated analysis methods[7]. The shared data containers support community-driven development and reproducible benchmarking of methods. The following sections provide an overview of some of the currently available tools, encompassing standard methods for taxonomic and other abundance tables as well as integrative analyses between modalities.

Community indicators. Various general indicators are available to quantify the microbial community state or other sample properties. Community diversity represents a common measure in microbial ecology[68] (Figure 5). Alpha diversity metrics generally quantify the observed or estimated number of distinct taxa (*richness*) and their distribution (*evenness*). Many diversity indices (*e.g.*, Shannon index), combine these two aspects. Phylogenetic diversity metrics additionally account the breadth of phylogenetic relatedness among community members[69], but are more computationally demanding than abundance-based metrics. However, a new implementation of the Stacked Faith's Phylogenetic Diversity improves computational efficiency by 1–2 orders of magnitude[70]. The framework supports analyses with rarefaction through the mia package[62]. Other global indices of community state include measures of dominance, rarity, skewness, kurtosis, modality, library size (number of reads), or absolute abundance quantification. Sample information in the TreeSE object (colData) can be enriched with these derived quantities. By default, the mia::addAlpha method returns a set of complementary alpha diversity measures[71]. Alpha diversity indices have been frequently applied to measure not only taxonomic diversity but also resistome diversity[72], and potentially other modalities.

Community similarity is commonly evaluated with beta diversity measures, which quantify multivariate dissimilarity between feature communities across all pairs of samples and can be used to compare community composition quantitatively. The framework supports common dissimilarity metrics in microbial ecology, including Bray-Curtis, Jaccard, Aitchison[61;68], and the phylogeny-aware UniFrac[73]. Community dissimilarity is commonly visualized with unsupervised ordination methods that project the data onto a lower-dimensional space[74]. Among unsupervised techniques, the most commonly used ones include Principal Component Analysis (PCA) and Principal Coordinate Analysis (PCoA), which encompasses both metric and non-metric multidimensional scaling (Figure 5). Statistical tests (*e.g.*, PERMANOVA) and supervised ordination methods, such as distance-based Redundancy Analysis (dbRDA), can complement the unsupervised analyses, and help uncover and quantify covariation between community composition and sample-specific covariates. This type of analysis can benefit from compositionally robust measures (*e.g.*, robust Aitchison distance[61]), or phylogeny-aware dissimilarity measures specifically developed for microbiome research (*e.g.*, UniFrac[73]). Sample dissimilarity provides the basis for many specialized microbiome techniques, including microbial community typing (clustering) with Dirichlet multinomial mixtures[75] and community state types[76], or factorization into community signatures[77]. These and other multivariate methods are provided by mia, miaViz, and additional single-cell packages (scater, scuttle[20]). Quantification of sample dissimilarities is a common task, and many omics modalities can benefit from shared, standardized methods provided by the integrative (Tree)SE framework.

Association analysis. Differential Abundance Analysis (DAA) and Differential Prevalence Analysis (DPA) methods are used to identify indicator microbial features whose abundances vary across study groups or continuous gradients, for example in terms of spatial or temporal distance between samples (Figure 5). Most DAA methods used in microbiome research support the (Tree)SE framework, such as ANCOM-BC2[78], LimROTS[23], MaAsLin3[79], and radEmu[80]. Leveraging the interoperability of this framework, several of these methods have also found applications with other omics modalities, including transcriptomics[81], metabolomics[79], and proteomics[23]. Importantly, recent studies have indicated that elementary methods, such as log-transformed Total Sum Scaled (TSS) counts coupled with a linear model, or presence/absence data with logistic regression, often yield the most consistent results[11;12]. Standard DAA typically focuses on individual features, ignoring interactions or other covariation among feature abundances. Thus, we support covariation analyses. While agglomerating collinear features as a preprocessing step can reduce multiple testing, increase statistical power, and improve interpretability, it is also prone to subjective parameterization and limited generalizability. Feature can be grouped based on clustering results from abundance profiles or on external information drawn from phylogenetic relations, taxonomic classifications or functional properties (*e.g.*, aggregating based on metabolite production[56]). Beyond these, early implementations of advanced methods, such as causal mediation analysis tailored to microbiome multi-omics data, have been implemented for SE[82], highlighting its readiness to support the adoption of constantly developing methodology.

Function- and sequence-level analyses. Taxonomic analysis is frequently complemented by functional analyses, either based on functional predictions derived from amplicon or metagenomic sequences, or from parallel metabolomic, metatranscriptomic or other functional assays. Our framework not only supports the incorporation of functional abundance tables within the same data objects, but many statistical methods implemented in the ecosystem are generic and can therefore be applied across different omics data types. Where appropriate, specialized methods are also available; for instance, regarding metabolome analyses, the notame package[83] has recently added SE support, and the R4MS project provides many interoperable tools[22]. The framework has found recent applications in extended analyses of metagenomic features, for instance, antimicrobial resistome composition[72]. We have also converted a large-scale compilation of resistome profiles in 8,972 metagenome samples[84] into the TreeSE format to support research and training activities in this area of microbiome analysis.

Spatio-temporal, ecological, and epidemiological models. Microbiome time series[85] encompass a wide range of study designs, ranging from short to long follow-up times, sparse to dense data, univariate to multivariate and multi-omic monitoring[86], and real, pseudo, or simulated time series. Moreover, an increasing number of studies incorporate spatial dimension in microbiome analysis[66]. While spatio-temporal analyses require specialized methods, the miaTime R package offers a set of data wrangling methods to support longitudinal, multi-modal microbiome analysis. In addition, Bioconductor contains applicable methods from other research domains, namely single-cell omics[20] and spatial transcriptomics[21], which use closely related data containers. Spatial and temporal information can be incorporated as a covariate (*e.g.*, location or time) or by estimating temporally or spatially resolved metrics, such as divergence (dissimilarity between consecutive time points), autocorrelation (similarity of closely related samples), and feature modality (whether a feature exhibits multiple abundance states). Another type of temporal analysis includes the time-to-event, or survival models, which are encountered regularly in epidemiological and population cohort studies; for microbiome data, these analyses require specialized approaches, as proposed in previous studies[87;88]. We also provide methods to simulate time series from standard models of microbial community dynamics including Hubbell's neutral model, generalized Lotka-Volterra, and the consumer-resource model through the miaSim package[89]. Our framework provides a robust basis for developing extended methods for longitudinal multi-omics, which is currently an active area of investigation[30].

Multi-modal integration. A key benefit of the ecosystem lies in its ability to support the integration of microbial abundance data with other omics modalities, such as resistomes[72], metabolomes[90], host genomes[91], transcriptomes, or proteomes[7;92] (Figure 5). As we have demonstrated, the integrative data containers support complex combinations of omics datasets[20;24;28]. By leveraging standardized multi-assay data structures, users can apply a growing number of integrative methods for multi-omics analysis. This eliminates the need for manual data wrangling, reduces the risk of errors (*e.g.*, sample mismatches), and enables the creation of modular, efficient, and reproducible workflows. Such integrative approaches in microbiome research can be broadly categorized into three groups: Machine Learning (ML) prediction, association-based approaches, and latent variable modeling[7;8;30;93–97]. ML approaches aim to integrate multiple omics modalities into a single predictive model. However, naive concatenation of feature tables can obscure modality-specific signals and reduce predictive performance. Methods such as IntegratedLearner[93] address this limitation by modeling each modality separately while leveraging their complementary predictive information within a unified framework. Association-based methods aim to quantify relationships between datasets. Whereas the cross-correlation analysis can be used to capture direct associations of samples or features between two datasets, anansi can supervise such analyses by incorporating known functional relations. While these association-based approaches are often easier to interpret, they do not explicitly account for the fact that microbes operate within complex communities. In contrast, latent variable methods such as multi-omics factor analysis[95], joint robust PCA[61] and multiblock partial least squares discriminant analysis[96] model shared latent structure across modalities, enabling the discovery and visualization of coordinated variation at the community level. The available methodologies are rapidly increasing, and we aim to accelerate the development, benchmarking, and standardization of multi-modal extensions for microbiome analysis through a systematic integrative framework proposed here.

Other methods and programming environments. Several other methods have been employed in microbiome data science. Examples include network analysis[98], ML-based prediction and classification[99], and microbe-set enrichment analysis[100;101]. The recent adoption of interoperable data containers and shared methodological approaches in related research fields can significantly expand the transfer of existing toolkits to adjacent or emerging disciplines. Importantly, this framework enables integration not only within the Bioconductor ecosystem but also with external pipelines, including bioBakery[51;102], and foreign platforms through language-specific interfaces, namely Rcpp for C++, reticulate for Python, and JuliaCall for Julia. These connections allow users to combine the most suitable tools across multiple frameworks, while ensuring cross-language interoperability of structured data through standard containers such as TreeSE and MAE.

Accessible analysis and community

Modern science is embodied by a critical community capable of assessing discoveries and replicating results[106]. Open source developer communities present a timely

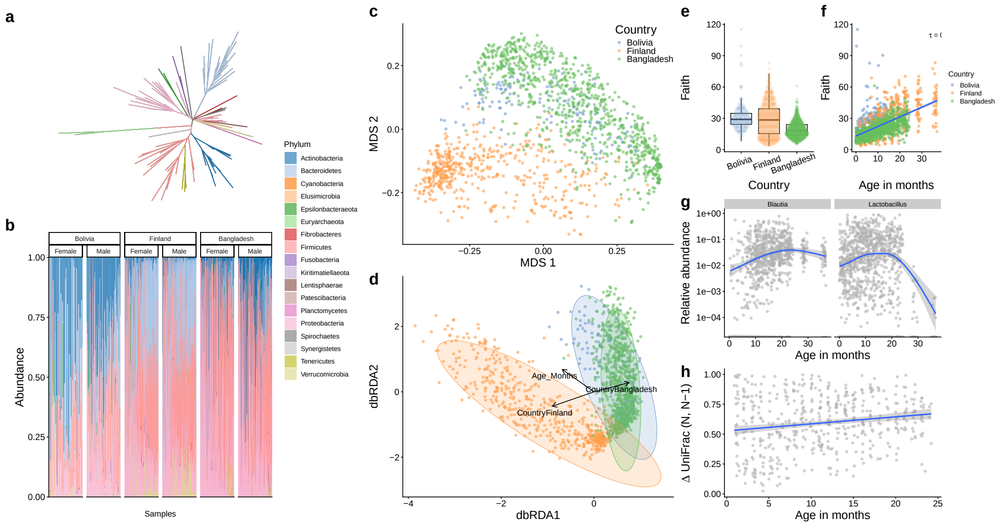

Figure 5: **Common visualization methods for microbiome data.** The selected visualizations demonstrate how the miaViz and scater packages effectively display various aspects of microbiome data, using the SprockettTHData time series dataset openly available via microbiomeDataSets[36]. This dataset consists of longitudinal 16S rRNA gut microbiome data from infant fecal samples collected in Finland, Bangladesh, and Bolivia[103–105]. **a**, Phylogenetic relationships. **b**, Phyla abundances of the samples ordered by age, stratified by country and gender. Actinobacteria is more abundant in Bolivia and Bangladesh compared to Finland. **c**–**d**, Multidimensional scaling using UniFrac distances and distance-based Redundancy Analysis (dbRDA) based on Aitchison distance, respectively. Both ordination methods shows clear pattern separating microbial profiles in different countries. dbRDA shows an additional association between microbial profile and age. **e**–**f**, Faith's phylogenetic diversity by country and age respectively. **g**, Differential Abundance Analysis (DAA) results. Two genera exhibit an association with age. **h**, UniFrac divergence between consecutive time points. Results indicate slow stabilization of microbial communities.

manifestation of this principle, with technical, legal, sociological and epistemic consequences[3]. FAIR principles for software support inclusive access to well-documented and user-friendly methods[1], and are facilitated by the extensive quality control and documentation standards of Bioconductor[24]. In addition to these technical demands, the development and quality control of research software significantly relies on a strong community engagement, which stimulates continuous software improvement and promotes the emergence and dissemination of evidence-based best practices[1;3;85]. We recommend starting with simple methods[11], calculating derived quantities such as diversity, prevalence, and other such measures early in the workflow, and complementing analyses with visual exploration. These practices can improve the transparency and interpretability of the analysis and its results. Expressive code and meaningful comments support reproducibility and verification.

Harmonized data structures and access to validated methods support standardized and transparent data wrangling across studies while facilitating method sharing, benchmarking, and reproducible comparisons. These capabilities help establish community best practices and build consensus while allowing users to flexibly select and evaluate methods appropriate for their specific data and research questions. The Bioconductor ecosystem naturally aligns with these goals through its open ecosystem of data resources, interoperable software, and educational materials. Community building is actively supported through online training resources and communication channels, making it easier for both developers and users to contribute to the continued development and standardization of the ecosystem. Together with interactive applications, these resources offer accessible entry points for developers and users, regardless of their programming experience.

The OMA book. *Orchestrating Microbiome Analysis with Bioconductor* (OMA) is an online book published at https://bioconductor.org/books/release/OMA that provides a comprehensive guide to the (Tree)SE framework for microbiome data science in Bioconductor, including practical guidelines, reproducible workflows, and training material for integrative microbiome analyses. The book is versioned according to the Bioconductor's biannual release cycle. The work incorporates rich feedback from the user community and microbiome training courses. The OMA book aligns to a broader Bioconductor convention, where prominent data structures[24;27;28] designed for multi-table data integration have been described across several domains, such as single-cell omics (OSCA book[20]) and spatial transciptomics (OSTA book[21]). These resources demonstrate how standard data containers can support statistical programming on interlinked multi-table datasets.

Interactive applications. Graphical user interfaces provide accessible entry points to the framework. The iSEEtree and miaDash packages provide interactive analysis and visualization tools for microbiome data[107]. The former extends iSEE to provide support for TreeSE objects, while the latter builds on iSEEtree to enable analyses without requiring any programming. Their interface supports automatic generation of source code and replicable workflows as well as analysis templates for users without a strong programming expertise. Moreover, the tidy R paradigm, available through tidyomics, simplifies the syntax and streamlines analytical workflows[108]. These tools allow users to analyze and visualize complex data combinations with ease.

Community. Its widespread adoption has made R one of the most cited software resources of our time[2] and the introduction of the shared data structures facilitates the broad interoperability of the methods ecosystem. Bioconductor has a strong tradition in fostering the R user community through shared data structures and training activities[5]. An active community of developers and users thrives online, supporting the software sustainability and promoting good practices in scientific computing[109]. The community interacts via several online communication channels, including Bioconductor Zulip, GitHub platform, and an email list. The development of the framework is influenced by feedback from the user community through these channels, regular meetings, and training workshops.

Outlook

To enjoy the benefits of automation while also maintaining flexibility, data scientists often recombine open-source software components into custom workflows. However, the lack of commonly adopted data representation standards can hinder the development, sharing, and application of modular, interoperable statistical workflows. Here, we present an R/Bioconductor ecosystem for multi-modal microbiome data science. Building upon previously established data structures[27;28], we have introduced a comprehensive, interoperable ecosystem developed as an extensive community effort. We also provide exhaustive guidance in the OMA book, thereby supporting independent learning and reproducible analysis in microbiome research and beyond.

The Bioconductor community is widely recognized for developing open research software for life sciences. Its strengths include minimal documentation standards, portability across operating platforms, quality control (*e.g.* peer-review, unit testing, and continuous integration), usability (APIs), robustness (dependency testing and deprecation mechanisms), and a biannual release cycle with semantic versioning. Over the past two decades Bioconductor has grown into a repository of more than 2,300 actively maintained software packages, of which 62 are now classified under the *Microbiome* Task View. At the core of the Bioconductor's data science strategy has been the idea of shared data representations[4;29]. This has enhanced interoperability, mitigated redundant efforts and facilitated the translation of methods between research domains[1;110]. In addition to these technical advances, the shared standards have played an important role in the formation of broad developer and user communities[5].

Microbiome research has shifted towards integrative analyses linking various omics towards enhanced mechanistic and causal insights[68;111]. Technological advances in long-read sequencing, functional assays, spatial profiling, and single-cell analysis have created an increasing demand for integrative analysis techniques that can link microbiome data with other omics along varying spatial and temporal dimensions. In concert with parallel developments in other omics, our framework specifically addresses the need to integrate and jointly analyze interlinked abundance tables and supporting hierarchical and tabular side information across different omics. Whereas our framework supports the basic data wrangling and microbiome data science operations, it also provides the structured input necessary for probabilistic inference and scalable computation. This is essential as the recent developments in microbiome and multi-omics workflows increasingly rely on advanced statistical and ML techniques such as Bayesian modeling and Gaussian processes to accommodate uncertainty and temporal or spatial structures[112]. Building upon these statistical foundations, the rapid emergence of foundation models and deep learning approaches in biomedicine calls for robust, scalable data representations[113] such as those we have adopted. Applications include transfer learning across studies, interpretable embeddings of microbiome multi-omic profiles, computational prediction of microbial function and metabolites[114;115], and integration with AI agents for hypothesis generation and automated analysis, among others[116;117].

Open data science infrastructures have become critical components of reproducible research and a vast body of applications critically rely on open-source methods and workflows[1;3;85]. Standardized data structures can enhance the transparency, reliability, and explainability of analysis workflows. We have documented the open data science ecosystem for multi-modal microbiome research built on widely adopted Bioconductor data containers, software libraries, data resources, tutorials, and a growing community of developers. Further extensions could include methods for temporal, spatial, and multi-omic analyses, improved support for community metadata standards and FAIR data practices (*e.g.*, Minimum Information about any (x) Sequence, MIxS [118]), along with an end-to-end case study demonstrating how an example analysis can be conducted in practice from raw data to results. Finally, supporting interoperability between R and other programming languages could facilitate the application of other ML frameworks. As the field evolves, the developer communities will continue to integrate new tools and approaches to improve integrative analyses of microbiome data.

Data availability

All datasets used in this article were made publicly available and can be accessed in R through the mia ecosystem [40].

Code availability

This article presented v1.1.1 of the OMA book (2026-05-07) published at https://bioconductor.org/books/release/OMA, which is dependent on R v4.6 and Bioconductor v3.24. Analytical scripts and related results of the benchmarking analysis were deposited on Zenodo [44].

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Author contributions

LL and TB initiated and coordinated the work. All authors contributed material, participated in manuscript preparation, and approved the final version.

Competing interests

The authors declare no competing interests.

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